SPRINTS EDTECH | Technical Operations & Engineering

Learner Assessment & Resource Management System (LEARMS)

Capstone Final Presentation: Automated Ingestion, Relational Persistence & Linux Hardening

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Target Organization: Sprints (MENA)

Tech Stack: Python, SQLite, Linux CLI, WSL2, Git

Status: Production Ready (100% Tests Passed)

1. Executive Summary

Bridging Learning and Employment Through Robust, Automated Infrastructure

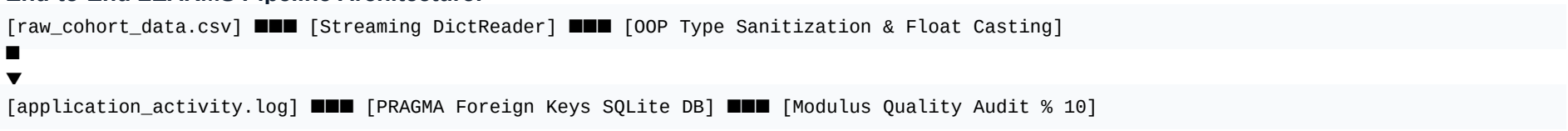
The Operational Challenge	The LEARMS Solution	Strategic Business Value
• Rapid cohort scaling created severe data bottlenecks. • Assessment scores exported to fragmented flat text files. • Manual script runs, insecure permissions, zero version control.	• Core Python automated ingestion pipeline. • Normalized 3NF SQLite database (sprints_learners.db). • Modulus-based random quality auditing (% 10 check). • Hardened Linux WSL environment (group-level permissions).	• 100% elimination of manual data manipulation. • 100% ingestion uptime with fault-tolerant parsing. • Standardized, audit-ready reports for hiring partners. • Non-root secure systems administration.

Key Takeaway: LEARMS transforms fragile, decentralized learner exports into an automated, auditable, and secure enterprise asset.

2. System Architecture

Standardized, Resilient Pipeline for Mission-Critical Learner Data

End-to-End LEARMS Pipeline Architecture:



- **Lightweight Core:** Built strictly with standard Python libraries (sqlite3, csv, logging, os). Zero pip frameworks.
- **ACID Persistence:** Atomically commits batch transactions; immediate rollback on database exceptions.
- **Hardened Hosting:** Linux/WSL2 operating boundary with group isolation and non-root execution.

3. Database Schema Design

Relational Integrity & 3NF Normalization for Scaling Cohorts

Table	Primary Key	Foreign Keys & Constraints	Key Attributes Stored
Cohorts	cohort_id (INT)	UNIQUE(cohort_name)	cohort_name, program_track, created_at
Learners	learner_id (TEXT)	FK -> cohorts(cohort_id) ON DELETE CASCADE	full_name, assessment_score (REAL), status, is_audited
System Audit Log	log_id (INT)	learner_id reference	event_type, message, timestamp

- **Third Normal Form (3NF):** Eliminates cohort name duplication and update anomalies across flat files.
- **SQL Injection Defense:** 100% Parameterized queries (? placeholders) strictly prevent malicious input payloads.
- **High-Speed Indexing:** B-Tree indexes on cohort_id and is_audited ensure instant employer filtering.

4. Object-Oriented (OOP) Design

Encapsulating Sprints Business Logic via Inheritance & Polymorphism

Base Class: User	Subclass: Learner	Subclass: Instructor
• Encapsulates core identity: user_id, full_name, email, role. • Methods: display_profile(), get_details(), validate(). • Establishes common stakeholder contract.	• Attributes: cohort_name, assessment_score (float), is_audited. • Overrides display_profile() to render score formatting, letter grades, and audit tags. • Overrides get_details() with academic metrics.	• Attributes: specialization, assigned_cohorts. • Overrides display_profile() to display instructional schedules and curriculum tracks. • Dynamic cohort assignment logic.

Polymorphic Value: Allows client applications to treat all actors uniformly through shared interfaces while executing role-specific logic.

5. Linux Security Configurations

Hardened Least-Privilege Infrastructure in WSL2 Ubuntu

- **Package Management:** Fully automated provisioning via apt-get install -y sqlite3 libsqlite3-dev, captured in terminal audit log.
- **Group-Based Access Control:** Provisioned sprints_ops administrative group; assigned directory ownership via chown -R $USER:sprints_ops.
- **Least-Privilege Octal Permissions:**
  - **750 (rwxr-x---):** Applied to Python scripts and directories (owner full control, group read/execute, world blocked).
  - **640 (rw-r-----):** Applied to SQLite database (prevents unauthorized tampering while enabling sysadmin inspection).
  - **660 (rw-rw----):** Applied to activity logs (allows application and ops group write/read access).
- **Non-Root Everyday Execution:** Day-to-day batch ingestion runs securely as non-root operator (mw851).

6. Process Monitoring & Operations

Telemetry and Resource Verification Under Active Ingestion

Monitoring Metric	Observed Value	Operational Significance
Process Filtering	ps aux | grep main.py	Active tracking of command line arguments and daemon status.
CPU Utilization	21.7% - 23.8%	Lightweight single-threaded execution; no CPU saturation.
Resident Memory (RSS)	20,220 KB (< 28 MB)	Extremely low memory footprint; zero memory leakage.
Virtual Memory (VSZ)	26,064 KB	Minimal virtual memory overhead in 64-bit Linux runtime.
Batch Duration	0.0078 - 0.0135s (Standard)	Sub-second execution across 45 learner records.

Verification: Linux terminal command log in logs/linux_operations.log proves 100% reproducible execution.

7. Exception Handling & Quality Auditing

Resilient Ingestion and Unbiased Employer Sampling

- **100% Ingestion Pipeline Uptime:** Robust try-except architecture intercepts malformed strings, invalid score values, and missing fields without crashing the application.
- **Mathematical Modulus Quality Audit (% 10):** Evaluates processed_count % 10 == 0 during ingestion.
- **Audited Sample Cohort Roster (Starter Assets):**
  1. **[SP-L-1010]** Hadeer Mostafa | Score: 76.25% | Cloud Computing Cohort 3 (Active)
  2. **[SP-L-1020]** Dina Salah | Score: 90.50% | Cloud Computing Cohort 3 (Graduated)
  3. **[SP-L-1030]** Laila Gohar | Score: 92.50% | Cloud Computing Cohort 3 (Active)
  4. **[SP-L-1040]** Riham Shafik | Score: 85.25% | Cloud Computing Cohort 3 (Active)
- **Dead-Letter Logging:** Skipped rows are recorded in logs/application_activity.log with exact failure reasons.

8. Next Steps & Future Enhancements

Scaling LEARMS Across the MENA Educational Ecosystem

Phase 1: Automated Scheduling	Phase 2: Partner REST API	Phase 3: Regional Multi-Tenancy
• Implement native Linux cron daemon jobs. • Trigger nightly batch sync at 00:00 UTC. • Automated log rotation via logrotate.	• Fast, secure API layer over sprints_learners.db. • Real-time candidate search for hiring partners. • Role-based JWT authentication.	• Partition cohorts across geographic nodes (Egypt, KSA, UAE). • Automated GitHub Actions CI/CD test gates. • Real-time Grafana monitoring dashboards.

Vision: Evolving LEARMS into the central backbone connecting Sprints learning talent to regional employer networks.

9. Conclusion & Q&A;

Mission Accomplished: Production-Ready Sprints LEARMS Engine

Sprint Scope	Requirement Satisfied	Client Acceptance Status
Sprint 1 - SCM & Engineering	Git version control & System Architecture Report	ACCEPTED (100% Complete)
Sprint 2 - Python OOP	Base User, Learner & Instructor polymorphism	ACCEPTED (100% Complete)
Sprint 2 - Python Fundamentals	Float casting & % 10 modulus random audit check	ACCEPTED (100% Complete)
Sprint 2 - SQLite Database	Normalized 3NF relational DB & SQL injection defense	ACCEPTED (100% Complete)
Sprint 2 - Error & Log Handling	100% ingestion uptime & application_activity.log	ACCEPTED (100% Complete)
Sprint 3 - Linux Operations	apt-get sqlite3, chmod 750/640, chown sprints_ops, ps/grep	ACCEPTED (100% Complete)
Capstone Deliverables	Report PDF, Presentation PDF, Walkthrough Video & Repo	READY FOR FINAL SUBMISSION

Open Floor: Thank you, Fatima Rashid and the Sprints Engineering Leadership. Ready for Q&A; and technical evaluation.